

V02.00.03

User Manual



1. iiOMA Helper installation



iiOMA

SW VERSION

02.00.03

TRACER FIRMWARE

-

Ready

Setting

Edger

Configuration

Calibration

Upgrade

CS Mode

Test Mode

Statistics

About us

● Folder unavailable

After download, run iioma-helper-setup.exe and accept the Windows admin prompt.

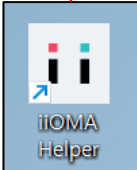
[Download helper](#)

Communication

Connect

Disconnect

Disconnected





iiOMA

SW VERSION

02.00.03

TRACER FIRMWARE

01.12.38

Loaded

Setting

Edger

Configuration

Calibration

Upgrade

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Test Mode

Statistics

About us

● Folder access available

Communication

Connect

Disconnect

Disconnected

- ① Access the iiOMA webpage, then click "Download Helper" to download and install it.
- ② If iiOMA Helper is not running, it cannot connect to the device or server.

2. Setting

Setting

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Communication

Connect

Disconnect

Connected — COM5

Save Path

Storage path

C:\Users\이호준\OneDrive\문서\새 폴더

...

Apply

When the helper is running, files are saved automatically at the path you enter. Type the full folder path, or use the "... " button to browse.

Data

Save Extension

dat

Data count

1440

Transfer

Upload the selected JOB file from the Edger tab to a remote destination. HTTPS POST runs natively in the browser. SFTP and FTP are not browser-native; configure a Bridge URL — your backend should accept the JSON payload and perform the upstream upload.

Protocol

HTTPS POST

SFTP

FTP

LMS push

LMS device push

Autocconnect receive in the Setting • LMS panel must be on for devices to appear here automatically. Press **Transfer** in the Edger tab to send job data directly to the selected device.

Target device

EDGER / LensEdger / TEST-1 — 192.168.0.3-50000

Refresh list

Clear list

Last announced at 2026-05-20 11:13 — helper will dial this Device on Transfer.

Remember credentials on this device (uncheck to wipe on tab close)

Idle

Test connection

LMS (DCS Host)

Host listening 0.0.0.0:33512

Devices connect to the host over TCP/IP to request job data. When the host is open below, the helper responds automatically. (DCS / OMA / VCA v3.13)

Enable LMS Host

ON

Bind address

0.0.0.0 (all interfaces)

TCP port (Remote)

33512

Autocconnect receive (UDP)

ON

Rendezvous port (UDP)

33511

Packet timeout (ms)

15000

Print communication log to console

ON

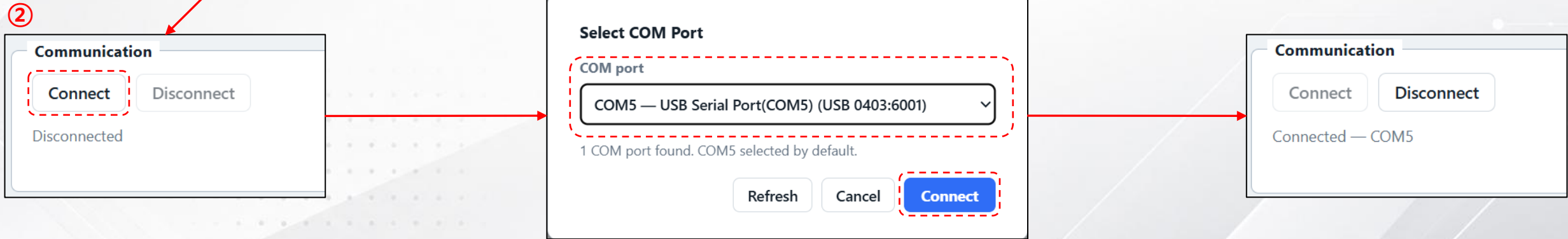
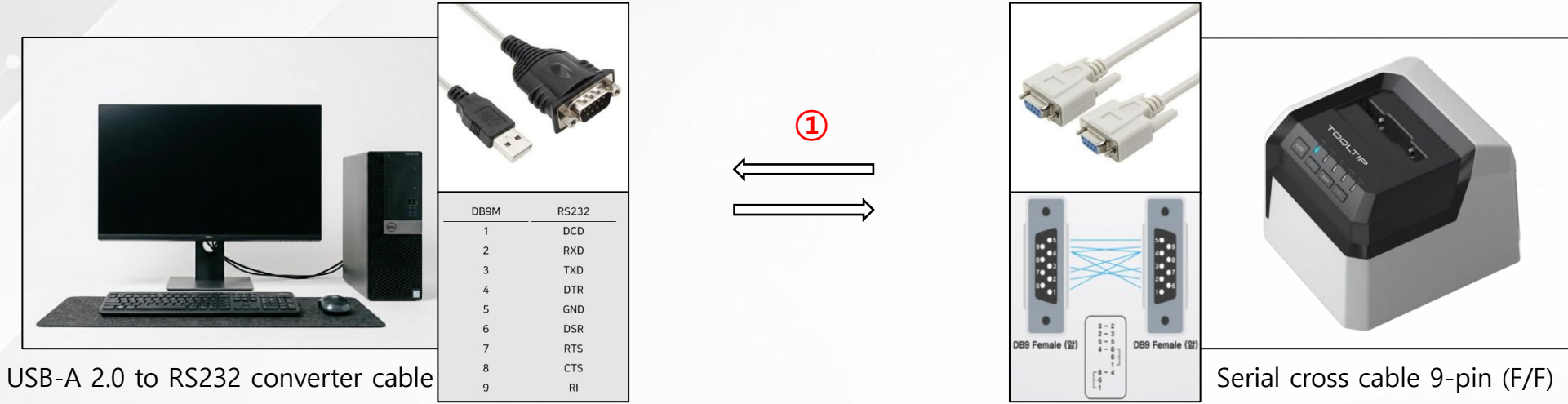
Default OMAV

3.13

SAVE

① The Setting tab is used to connect iiOMA to the tracer and to connect iiOMA to the edger and server using various communication methods.

2. Setting (Communication)



① Connect the PC and the tracer using the following cable. (Please connect it to the **[PC] port** of the tracer.)

If a cable with a different pin map is used, the connection may not work.

② Connect the cable, check and select the connected COM port on the PC, then press Connect to complete the connection.

2. Setting (Save Path / Data)

1 Save Path

Storage path

Full path — e.g. C:\Users\YourName\Documents\jobready

...

Apply

When the helper is running, files are saved automatically at the path you enter. Type the full folder path, or use the "..." button to browse.

① Press [...] to select the folder where the JOB Data will be saved, or enter the folder path directly, then press [Apply].

Data

Save Extension

job

▼

Data count

1440

▼

- ② Select the file extension for saving the JOB Data. (dat / txt / oma / job)
- ③ You can change the file shape resolution (number of points) of the generated data.

2. Setting (Transfer_HTTPS POST)

Transfer

Upload the selected JOB file from the Edger tab to a remote destination. HTTPS POST runs natively in the browser. SFTP is not browser-native; configure a Bridge URL — your backend should accept the JSON payload and perform the upstream upload.

Protocol

☒ HTTPS POST
 ☐ SFTP
 ☐ LMS push

HTTPS POST settings

1

URL

https://service.iineer.com/api/public/job-preview/parse

2

Method

POST

Form field name

file

3

Bearer token (optional)

eyJ...

4

Extra headers (one "Key: Value" per line)

X-Site: clinic-01

- ① Enter the URL of the server that will receive the JOB file via the HTTPS protocol.
- ② Specify the HTTPS communication method and the Form field name. (The default values are POST and file.)
- ③ The Bearer Token is optional and can be used to enter an Authorization token.
Bearer is automatically prefixed to the token value.
- ④ Additional headers required by the receiving server can also be added.
Headers should be entered in key:value format, with each header separated by a new line.

2. Setting (Transfer_SFTP)

Transfer

Upload the selected JOB file from the Edger tab to a remote destination. HTTPS POST runs natively in the browser. SFTP is not browser-native; configure a Bridge URL — your backend should accept the JSON payload and perform the upstream upload.

Protocol

☐ HTTPS POST
 ☒ SFTP
 ☐ LMS push

SFTP settings

① Bridge URL (server-side relay)

https://relay.example/iioma/sftp

② Host

Port

22

③ Username

iioma_upload

Password

.....

④ Remote path

/upload

- ① Please enter the Bridge URL if available. If not, this field may be left blank.
- ② Please enter the host address for SFTP communication. (Public IP or Public DNS),
Please enter the port number used by the SFTP server for incoming connections.
- ③ Please enter the Username and Password used to authenticate with the SFTP server.
These credentials will be used to establish a secure connection for file transfer.
- ④ Please enter the path of the remote folder where the JOB files will be uploaded.
The uploaded files will be transferred and stored in the specified directory on the SFTP server.

2. Setting (Transfer_LMS push)

Transfer

Upload the selected JOB file from the Edger tab to a remote destination. HTTPS POST runs natively in the browser. SFTP is not browser-native; configure a Bridge URL — your backend should accept the JSON payload and perform the upstream upload.

Protocol

☐ HTTPS POST
☐ SFTP
☒ LMS push

LMS device push

Autoconnect receive in the Setting ▶ LMS panel must be on for devices to appear here automatically. Press **Transfer** in the Edger tab to send job data directly to the selected device.

Target device

①

EDGER / LensEdger / TEST-2 — 192.168.0.19:50000

▼

②

Refresh list

Clear list

Last announced at 2026-05-27 16:01 — helper will dial this Device on Transfer.

☒ Remember credentials on this device (unchecked to wipe on tab close)

④

LMS target ready: 192.168.0.19:50000

③

Test connection

- ① The LMS Push function allows JOB data to be transmitted to ophthalmic devices (such as Edgers) connected to the same local network. Please select the target device that will receive the JOB data.
- ② If the desired device does not appear in the list, click the Refresh button to rescan and update the available device list.
- ③ Once a device has been selected, you can use the Test Connection function to verify network communication with the selected device.
- ④ The connection result and status will be displayed for confirmation.

2. Setting (LMS_DCS Host)

LMS (DCS Host)

Host listening 192.168.0.15:33512

Devices connect to the host over TCP/IP to request job data. When the host is open below, the helper responds automatically. (DCS / OMA / VCA v3.13)

- Enable LMS Host

ON
- Bind address

[wifi] Wi-Fi — 192.168.0.15

TCP port (Remote)

33512
- Autoconnect receive (UDP)

ON
- Rendezvous port (UDP)

33511

Packet timeout (ms)

15000
- Print communication log to console

ON
- Default OMAV

3.13

- ① To use the LMS Push function, this toggle must be turned **ON**.
- ② You can specify the host IP address to be used for communication. Please select the Wi-Fi network interface.
If no Wi-Fi interface is detected, the system will bind to 0.0.0.0 instead. Please specify the TCP port to be used for communication. The officially recommended default value is 33512.
- ③ To listen for UDP broadcast packets, this toggle must be turned **ON**.
- ④ Please specify the UDP port to be used for broadcast communication.
The officially recommended default value is 33511. You can also configure the packet timeout value.
The default timeout is 15000 ms.
- ⑤ You can choose whether communication logs should be displayed during operation.
- ⑥ Please select the version of the DCS OMA protocol to be used for communication. Default is 3.13.

3. Edger

Setting

Edger

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Folder access available

List

JOB 4 files

#20260521_001.job

#20260521_003.job

#20260521_002.job

TEST #1.job

CREATED 2026-05-21 16:55

MODIFIED 2026-05-21 14:07

Data Manager

DESTINATION

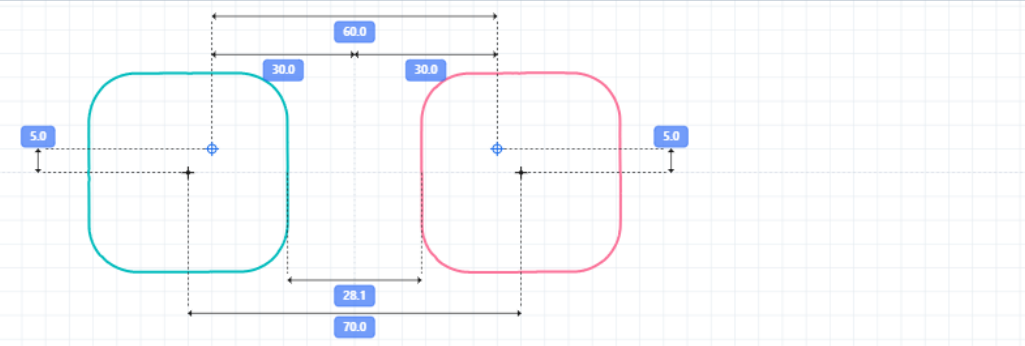
HTTPS POST - https://service.ineer.com/api/public/job-preview/parse

Lens

#20260521_001.job

R-L

L-R



SPH 0.00

ADD 0.00

CVL 0.00

PRVM 0.00

Axis 0.00

PRVA 0.00

Curve 1.14

Frame

ZTILT 0.03

SPH 0.00

ADD 0.00

CVL 0.00

PRVM 0.00

Axis 0.00

PRVA 0.00

LENS MATERIAL

☒ PL ☐ PC ☐ HI ☐ GL ☐ TR

BEVEL POSITION

☒ Percent ☐ Front ☐ Back ☐ Curve

50

%

FRAME TYPE

☐ Bevel ☐ Flat ☐ Mini-bevel ☒ Groove

☐ Polish

Mini Height

Width

Height

0.60

0.60

0.40

SIZE ADJUSTMENT

☒ HBOX / VBOX ☐ CIRC ☐ CIRC3D

SIZE ADJUSTMENT

Delta

R / L

0

Right

41.90

41.83

149.96

149.97

Left

41.90

41.93

149.98

149.99

Order Info

Job Name

#20260521_001

Prescription No.

Account No.

Last Name

First Name

Address

Phone

Memo

Apply

Transfer

Order Preview

① **List** : List of JOB Data stored in the configured folder. The JOB Data read by the tracer is saved here.

② **Lens** : This is the image of the frame read by the tracer.

You can change the processing settings or values before sending the data to the Edger or server.

③ **Size Adjustment** : You can adjust the lens size by selecting HBOX / VBOX / CIRC / CIRC3D.

④ **Order Info** : You can enter customer information.

iNEER
iNEER CO., LTD.



INEER
INEER CO., LTD.

3. Edger (Lens)

① Size design area
(PD / IPD / OH / DBL /MPD)

①

Smooth

Grid ☒

②

SPH	CYL	Axis
0.00	0.00	0.00
ADD	PRVM	PRVA
0.00	0.00	0.00

③

Curve	Frame	ZTILT
4.60		3.22

②

SPH	CYL	Axis
0.00	0.00	0.00
ADD	PRVM	PRVA
0.00	0.00	0.00

④

LENS MATERIAL

☒ PL ☐ PC ☐ HI ☐ GL ☐ TR

⑤

BEVEL POSITION

☒ Percent ☐ Front ☐ Back ☐ Curve

50

%

⑥

FRAME TYPE

☒ Bevel ☐ Flat ☐ Mini-bevel ☐ Groove

☐ Polish

Mini Height

Width

Height

0.60

0.60

0.40

④ You can select the lens material.

⑤ You can select the bevel and grooving positions.

⑥ You can select the Frame Type and adjust the dimensions.

② You can enter the values for the right lens.

② You can enter the values for the Left lens.

③ The curve and angle are displayed.

3. Edger (Size Adjustment)

① You can select the area where the size will be adjusted.

SIZE ADJUSTMENT

☐ HBOX / VBOX ☐ CIRC ☒ CIRC3D

	SIZE ADJUSTMENT		FRAME SIZE	
Delta	R / L		Right	Left
HBOX	0		50.95	51.10
VBOX	0		48.61	48.51
CIRC	0		158.19	158.13
CIRC3D	0		158.22	158.16

② A blue outline appears around the selected area, and you can enter the dimensions to adjust.

③ Displays the size of the selected JOB Data.

3. Edger (Order Info)

①

Order Info

Job Name

ii20260520_001

Prescription No.

Account No.

Last Name

First Name

Address

Phone

Memo

②

Apply

③

Transfer

④

Order Preview

①

You can enter and save detailed customer information for each JOB.

②

Applies the changes such as size adjustments and processing settings.

③

Transfers the JOB data to the device or server configured in the Setting tab.

④

After entering the customer information and pressing the Order Preview button, the following window appears, allowing you to print or save it as a PDF file.

④

iNEER

ORDER REPORT

Metal 001

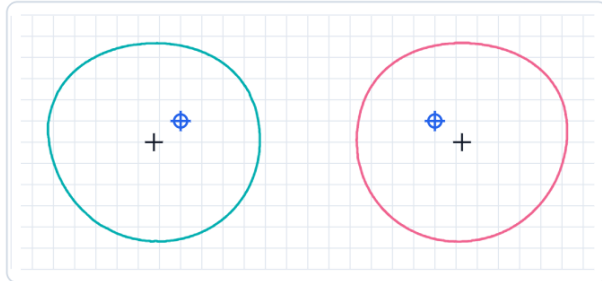
Printed

May 20, 2026, 5:55 PM

File

Metal 001.dat

LENS SHAPE



ORDER / CUSTOMER

Job Name	Metal 001
Prescription No.	-
Account No.	-
Last Name	-
First Name	-
Address	-
Phone	-

TRANSFER

File Created	May 20, 2026, 5:05 PM
Transfer History	LMS May 20, 2026, 5:05 PM LMS push - EDGER / LensEdger / TEST-2 - 192.168.0.19:50000

RX / OPTICAL LAYOUT

OPTICAL LAYOUT

	Right	Left
PD		60.00 mm
MPD		72.68 mm
DBL		22.83 mm
IPD	30.00 mm	30.00 mm
OH (Box Center)	5.00 mm	5.00 mm

FRAME / SHAPE

Frame Curve	2.88	Frame Tilt	3.40 deg
		Right	Left
Horizontal Box	50.01	49.69	
Vertical Box	46.77	46.87	
Circumference	153.01	152.96	
3D Circumference	153.02	152.97	

PRESCRIPTION

	Right	Left
Sphere	0.00	0.00
Cylinder	0.00	0.00
Axis	0.00	0.00
ADD	0.00	0.00
PRVM	0.00	0.00
PRVA	0.00	0.00

EDGING / MATERIAL

Lens Material	Edge Type	Polish	Bevel Position	Bevel Value
PL	Bevel	No	Percent	50 %

MEMO

4. Configuration

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General

Beep Sound
Open Incoming Data
Job Name Format
Pattern Default Design
Design Sync

ii{YYYY}{MM}{DD}_{SEQ3}

Groove
Bevel

Design

PD
DBL
OH Reference
OH Box Center
OH Eye Bottom
OH Lens Bottom

60.00
17.00
OH Box Center
5.00
20.00
20.00

Layout

Lens material
Frame Type
Bevel Position

PL
PC
HI
GL
TR

Bevel
Mini Bevel
Flat
Groove

Percent
Front
Rear
Base Curve

Groove Position

50.00
1.00
1.00
2.00

Percent
Front
Rear
Base Curve

Size Correction

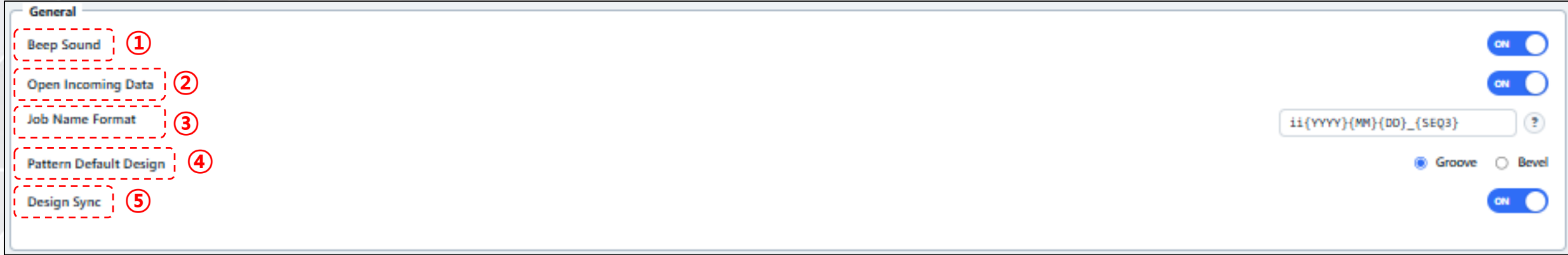
NONE
CIRC
CIRC3D
HVBOX

	Plastic	Metal	Pattern	Optyl
CIRC	0.00	0.00	0.00	0.00
CIRC3D	0.00	0.00	0.00	0.00
HBOX	0.00	0.00	0.00	0.00
VBOX	0.00	0.00	0.00	0.00

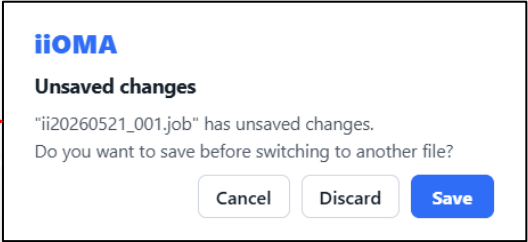
SAVE

- ① **General** : General Settings.
- ② **Design** : Design Settings.
- ③ **Layout** : Set the default layout when loading data from the tracer.
- ④ **Size Correction** : For each frame material, you can select CIRC / CIRC3D / HBOX / VBOX to preset the size adjustment for the lens data read by the tracer.

4. Configuration (General)



- ① **Beep Sound** : You can turn on or off the alarm sound.
- ② **Open Incoming data** : When set to "ON", if data is received from the tracer while editing job data, the following window appears and the corresponding job data is displayed on the screen.
- ③ **Job Name Format** : You can change the format of the saved JOB name.
- ④ **Pattern Default Design** : You can specify the pattern of the data to be loaded by selecting Bevel/Grooving.
- ⑤ **Design Sync** : When ON, the IPD design value is applied equally to both left and right.



4. Configuration (Design)

Design

PD ①

DBL ②

OH Reference ③

OH Box Center

OH Eye Bottom ④

OH Lens Bottom

60.00

17.00

OH Box Center ▼

5.00

20.00

20.00

- ① **PD** : You can preset basic value of PD (Pupillary Distance) when job data is open.
- ② **DBL** : You can preset basic value of DBL (Distance between lenses) when job data is open.
(Basic DBL value is only applied when you did not enter DBL value.)
- ③ **OH Reference** : You can select the OH reference of the JOB data from
OH Box Center / OH Eye Center / Lens Bottom Center.
- ④ **OH Value** : Default value when selecting the OH reference standard.

OH Box Center ▼

OH Box Center

OH Eye Bottom

OH Lens Bottom

4. Configuration (Layout)

Layout

① Lens material

☒ PL
 ☐ PC
 ☐ HI
 ☐ GL
 ☐ TR

② Frame Type

☒ Bevel
 ☐ Mini Bevel
 ☐ Flat
 ☐ Groove

③ Bevel Position

☐ Percent
 ☐ Front
 ☒ Rear
 ☐ Base Curve

Groove Position

☒ Percent
 ☐ Front
 ☐ Rear
 ☐ Base Curve

① **Lens material** : You can set the material of the data to be loaded.

② **Frame Type** : You can set the frame type of the data to be loaded.

③ **Bevel&Groove Position** : You can set the processing position and numerical values for beveling and grooving.

4. Configuration (Size Correction)

Size Correction

☐ NONE

☒ CIRC

☐ CIRC3D

☐ HVBOX

	Plastic	Metal	Pattern	Optyl
CIRC	0.00	0.00	0.00	0.00
CIRC3D	0.00	0.00	0.00	0.00
HBOX	0.00	0.00	0.00	0.00
VBOX	0.00	0.00	0.00	0.00

- ① Size Correction select

Select the method to use for Size Correction.
- ② Material

If you enter Size Correction values for each frame material, the entered values will be automatically applied when the data is loaded.

5. Calibration

Setting | Edger | Configuration | Calibration | Upgrade | CS Mode | Test Mode | Statistics | About us

● Folder access available

Calibration Target

☒ Stylus Shaft

☐ Frame

☐ Pattern

Start

Stop

Tracing

Result

Right CIRC

Left CIRC

- ① **Stylus Shaft** : Calibration to optimize the reading process by measuring the sensitivity and weight of the stylus shaft.
A separate jig is not required.
- ② **Frame** : Install the frame JIG and perform the calibration. When the calibration is complete, the circumference value will appear in the "Result" section. Please check whether it shows PASS.



Result

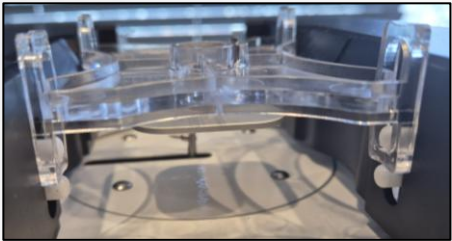
Right CIRC

149.92 (PASS)

Left CIRC

149.94 (PASS)

- ③ **Pattern** : Install the pattern jig and perform calibration. Once calibration is complete, the circumference value will be displayed in the "Result" section. Check if it says "Pass".



Result

Right CIRC

149.99 (PASS)

Left CIRC

149.99 (PASS)

- ④ **Tracing** : Button for reading only to check the perimeter value after calibration.

6. Upgrade

Setting Edger Configuration Calibration **Upgrade** CS Mode Test Mode Statistics About us

● Folder access available

Firmware

Server **Online** Refresh list

Firmware file

tracer_v01.12.37.hex[SERVER] Browse local... **DOWNLOAD**

Turn off the Tracer, then turn it back on while holding the TYPE button for 3 seconds. If all LEDs are lit, the device is ready for the upgrade. Then, press the UPGRADE button.

Ready

- ① When entering the Upgrade tab, the latest version marked with [SERVER] is automatically selected.
- ② Press Download, and once the download is complete, the file will be saved to DISK and the button will change to UPGRADE.

Firmware file

tracer_v01.12.37.hex[DISK] Browse local... **UPGRADE**

Turn off the Tracer, then turn it back on while holding the TYPE button for 3 seconds. If all LEDs are lit, the device is ready for the upgrade. Then, press the UPGRADE button.

Download complete. Click UPGRADE to flash.

- ③ Turn off the Tracer, then turn it back on while holding the TYPE button for 3 seconds. If all LEDs are lit, the device is ready for the upgrade. Then, press the UPGRADE button.

7. Test Mode



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Auto Refresh

OFF

All Initialize

Test Mode	Sensor	Value	Movement
X Axis	PI	Open 0.0	Initialize Open Close
Theta Axis	PI	Open 0.0	Initialize Open Close
R Encoder		16	Initialize Push Pull
Z Encoder		1	Initialize Up Down
Gripper	PI	Close	Initialize Grip Release
MIN	PI	Open	
MID	PI	Open	
MAX	PI	Close	

Status: **Connected** — COM5

Connect COM

Disconnect

Stop

① This screen is the test mode, allowing you to check the status of functions and sensors.

8. Statistics

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● Folder access available

①

Statistics

Total 46 | Right 23 | Left 23

②

Type	Plastic	Metal	Optyl	Pattern	Demo	Sum
Right	0	1	0	0	0	1
Left	0	1	0	0	0	1
Both	14	16	14	0	0	44

- ① **Statistics** : Displays the number of times the frame has been read by the tracer.
- ② **Material** : Displays the number of times the tracer has read the right, left, and both sides according to the frame material.

[Setting](#) [Edger](#) [Configuration](#) [Calibration](#) [Upgrade](#) [CS Mode](#) [Test Mode](#) [Statistics](#) [About us](#)

● Folder access available

iiNEER Corp.
Cutting edge technologies for opticians

We, at iiNEER Corp., strive to provide cutting edge technology to opticians including world's first web-based remote customer service platform. iiNEER Corp. is established to provide more convenient and valuable service that impresses customers with our unique technology. With years of professional experience and competitive patents, we, iiNEER will lead the optical lens processing industry.

[!\[\]\(b93c3e1add16fe46100bba7a6da1e82f_img.jpg\) Official Website](#)
[!\[\]\(e03fe9a24fe5105c8f59e912ae7b8724_img.jpg\) Official YouTube Channel](#)
[!\[\]\(b4572a044582c68c9e6e6b6b9b95c325_img.jpg\) Download Manual \(PDF\)](#)
[!\[\]\(a4848a7f290c3fd14bf2276a5b09747a_img.jpg\) Privacy Policy](#)

- ① You can access the official iineer website and YouTube channel through the posted links.
- ② You can download the manual.
- ③ You can check the Privacy Policy.